II Jacobson Radical Theory

4 The Jacobson Radical

Exercise 1.1: secretly Lemma 4.1 part a)

Let R be a ring. Define the Jacobson radical $J(R)$ as the intersection of all maximal left ideals. Prove that $J(R)$ is, in fact, a two-sided ideal. (As a lemma, prove that an element $x \in R$ belongs to the Jacobson radical if and only if $1 - axb$ is invertible for all $a, b \in R$.)

Exercise 2.1: secretly Lemma 4.1 part b)

Let R be a ring and let $r \in R$. Prove that r belongs to $J(R)$ iff $rM = \{0\}$ for all simple left ${}_RM$.

Exercise 3.1: secretly Theorem 4.14 part a)

A ring is said to be **semiprimitive** if $J(R) = \{0\}$. Prove that if R is a semisimple ring, then R is artinian and semiprimitive.

Proof.

Suppose R is semisimple. Then, there exists some left ideal I such that:

$$R \;=\; J(R) \oplus I$$

If $J(R) \neq (0)$, then $I \neq R$, and is thus contained in some maximal ideal I'. But then $I' \nsubseteq J(R)$, a contradiction. Thus, R must be semiprimitive. Since a ring is semisimple if and only if it is noetherian and artinian, we are done. ■

Exercise 4.1

Find $J(\mathbb{Z}_n)$ for all $n \in \mathbb{N}$.

Proof.

Consider the maximal ideals of $\mathbb{Z}_n$. If $\gcd(m, n) = 1$, then in $\mathbb{Z}_n$, $(m) = \mathbb{Z}_n$.

If $d \mid n$ and d is not prime, then $d = p^k$ for some prime p and $k > 1$. But then $(d) = (p^k) \supseteq (p)$, so (p^k) is not maximal for any $k > 1$.

Thus, the maximal ideals of $\mathbb{Z}_n$ are (p) for all prime $p \mid n$. That is, if we write $n = p_1^{m_1} \cdots p_k^{m_k}$, then the maximal ideals of $\mathbb{Z}_n$ are (p_i) for all i.

Recall that for ideals of $\mathbb{Z}$, we have that:

$$(a) \cap (b) \;=\; (\operatorname{lcm}(a, b))$$

Thus:

$$J(\mathbb{Z}_n) \;=\; \bigcap_{i=1}^{k} (p_i) \;=\; (\operatorname{lcm}(p_1, \ldots, p_k)) \;=\; (p_1 \cdots p_k)$$

$\blacksquare$

Exercise 5.1

Let R, S be two rings. What is $J(R \times S)$?

Proof.

dupe of problem 8 below? ■

Exercise 6.1

Let R be a ring and let $n \geq 1$. What is $J(M_n(R))$?

Exercise 7.1

Let R be a ring with ideal I. What is $J(R/I)$?

Exercise 8.1: Exercise 4.12B

Let A, B be two rings. Show that $J(A \times B) = J(A) \times J(B)$.

Exercise 9.1: Exercise 4.21

Let $\mathrm{GL}_n(R)$ denote the group of units in the matrix ring $M_n(R)$. Suppose that I is an ideal of R contained in $J(R)$. Show that the map $\mathrm{GL}_n(R) \to \mathrm{GL}_n(R/I)$ is onto.

5 Jacobson Radical under Change of Rings

Exercise 10.1: Corollary 5.2

Let R be a commutative ring. Prove that the polynomial ring $R[x]$ is semiprimitive (meaning its Jacobson radical is zero) if and only if R is a reduced ring (meaning it has no nilpotents).

Exercise 11.1: Theorem 5.3

Prove that if k is a field and R a finitely-generated k-algebra, then R is semiprimitive.

Exercise 12.1: Proposition 5.6

Let R be a ring, and let A be a subring of R. Suppose that A is a direct summand of R (as a left R-module). Prove that $J(A) = J(R) \cap A$.

Exercise 13.1

Let R be a commutative ring. Prove that the intersection of all prime ideals of R is equal to the set of nilpotents in R.

Exercise 14.1

Let R be a commutative ring and let $p(x) = \sum_{i=0} a_i x^i$ be in $R[x]$. Prove that $p(x)$ is a unit in $R[x]$ if and only a_0 is a unit in R, and $a_1, \ldots, a_n$ are nilpotents in R.

Hint: show that $a_1, \ldots, a_n$ belong to every prime ideal of R. Thus use the previous exercise to deduce that $a_1, \ldots, a_n$ are nilpotent.

Exercise 15.1: Exercise 5.6

Let R be a ring, let $R[[t]]$ be the ring of formal power series in an indeterminate t with coefficients from R. Note that $tr = rt$ for all $r \in R$. Let (t) denote the ideal of $R[[t]]$ generated by t. This coincides with the set of power series with no constant term:

$$(t) := \{tf(t) : f(t) \in R[[t]]\} = \{f(t) \in R[[t]] : f(0) = 0\}$$

Prove that the Jacobson radical of $R[[t]]$ is:

$$J(R[[t]]) = J(R) + (t)$$

Exercise 16.1: secretly Theorem 4.14 part b)

Prove that if R is artinian and semiprimitive, then R is semisimple.

Proof.

Suppose R is artinian and semiprimitive. Note R must have the following two properties:

a) Every nonzero left ideal U contains a minimal left ideal B. This follows from the fact that R is artinian.

b) Every minimal left ideal B is a direct summand of ${}_RR$ with a maximal left ideal.

To see property b), note that since $J(R) = 0$, then there necessarily exists a maximal left ideal m such that $B \not\subseteq m$. Since $B \cap m$ is an ideal of B and B is minimal, then $B \cap m = (0)$. Since m is maximal, note that $B + m = {}_RR$, and therefore ${}_RR = B \oplus m$.

Now, take any minimal left ideal B_1 and, by b), write:

$$ {}_RR = B_1 \oplus U_1 $$

Then $U_1 \neq 0$, so by a), there exists a minimal left ideal $B_2 \subseteq U_1$. By b), B_2 is a direct summand of ${}_RR$ and thus U_1, so we can write:

$$ U_1 = B_2 \oplus U_2 \implies {}_RR = B_1 \oplus B_2 \oplus U_2 $$

Continuing ad infinitum, if R is not semisimple, then we necessarily always have that $B_{k+1} \subsetneq U_k$. In other words, we get an infinite descending chain:

$$ U_1 \supsetneq U_2 \supsetneq U_3 \supsetneq \cdots $$

This contradicts the fact that R is artinian, so we must have that R is semiprimitive as needed. ■